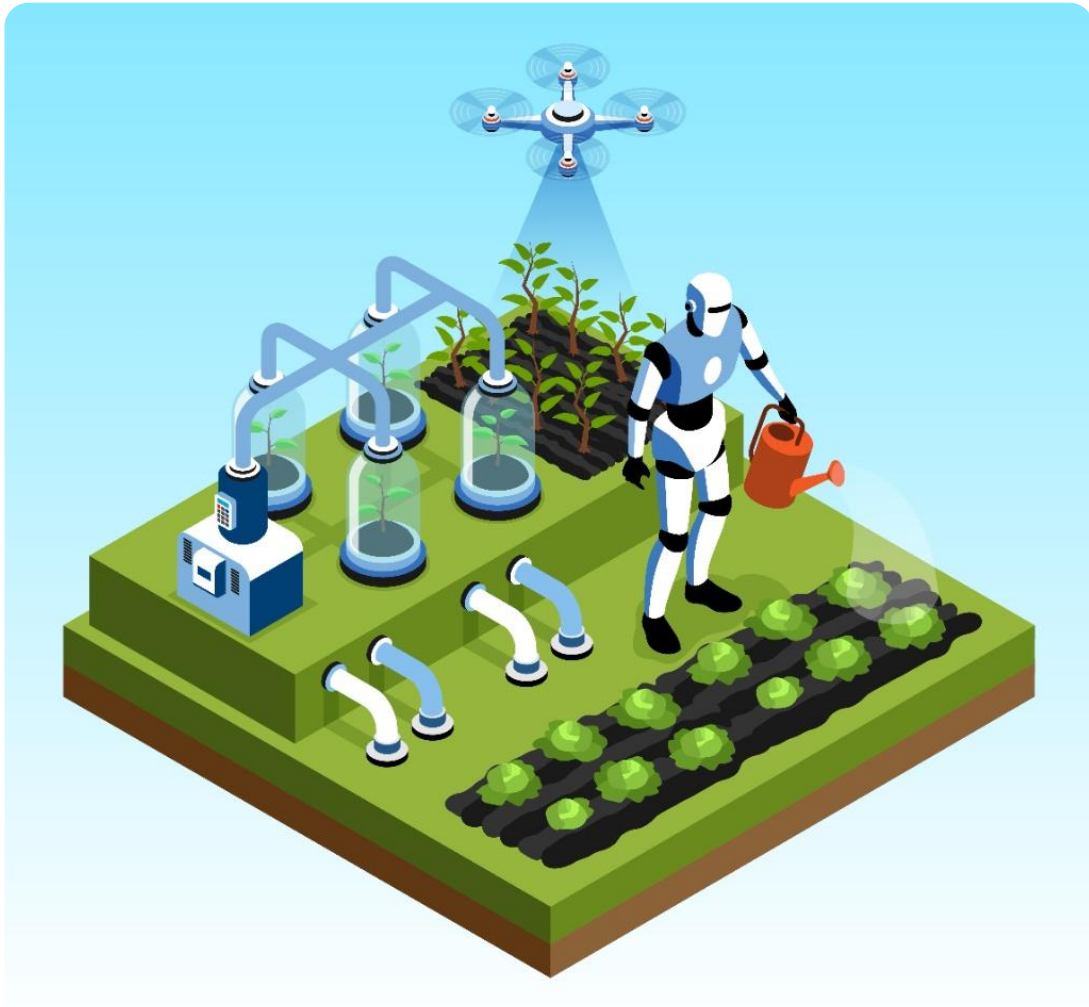


AGRIBOT

Agriculture Crop Recommendation System



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COLLEGE: Kendrapara Autonomous College

PROJECT

NAME: Agribot

PROJECT TITLE: Agriculture Crop Recommendation System

Introduction

It is an online Platform that helps farmers to make informed decisions by recommending suitable crops based on soil and weather conditions. The platform also provides an online marketplace where farmers can buy seeds and fertilizers and sell harvested vegetables directly to Godown. Customers can order fresh vegetables.

Problem Statement & Context

Problem Statement

- Low crop yield and profit.
- Lack of expert agricultural guidance.
- Difficulty accessing quality seeds and fertilizers.
- Dependence on middlemen for selling vegetables.
- No integrated platform for farming needs.
- Wrong crop selection by farmers.

Why does this Problem Exist?

- Lack of soil and weather analysis.
- Limited access to agricultural information.
- Traditional farming practices.
- Poor market connectivity.
- Lack of digital solutions in agriculture.
- Insufficient data-driven decision-making.

Stakeholders & Objectives

Stakeholders

Those who have some interest over this platform:

- **Primary Stakeholders:** These stakeholders directly use or benefit from the system.
 - Farmers
 - Buyers/Consumers
 - Seed Suppliers

- Fertilizer Suppliers
- **Secondary Stakeholders:** These stakeholders support or manage the system.
 - System Administrator
 - Delivery Partners
 - Agricultural Experts
- **External Stakeholders:** These stakeholders are outside the system but have an interest in its outcomes.
 - Government Agricultural Departments
 - Agricultural Organizations/NGOs
 - Financial Institutions (for agricultural loans and payments)

Objectives

- Recommend the most suitable crops for cultivation.
- Enable online purchase of seeds and fertilizers.
- Allow farmers to sell vegetables through the platform.
- Improve agricultural productivity and profitability.
- Support data-driven farming decisions.

Main Modules

1. Crop Recommendation Module

Inputs:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- pH
- Rainfall rate

Outputs:

- Recommended Crop
- Suitability Percentage

- Crop Cultivation Tips

2. Seed Store Module

- Browse seeds
- Search seeds
- Add to cart
- Place orders
- Online payment

3. Fertilizer Store Module

- View fertilizer catalog
- Compare products
- Purchase fertilizers
- Track orders

4. Vegetable Storage Module

- Farmers list vegetables for sale
- Storage management browse products
- Order management
- Delivery tracking

6. Admin Module

- Manage users
- Manage crops
- Manage seeds and fertilizers
- Manage orders
- Generate reports

7. Farmer Module

- Register and log in
- Enter soil and weather details
- Receive crop recommendations
- Buy seeds and fertilizers
- List vegetables for sale

- Track orders and sales

8. Buyer Module

- Register and log in
- Browse vegetables
- Place orders
- Make payments
- Track deliveries

Users & Activities

Users: Who will interact with the system

- Farmer
- Customer/Buyer
- System Admin
- Godown Manager
- Agricultural Experts
- Delivery person

Activities: What are done by the users

System Admin:

- Log in
- Manage users
- Approve Farmer Registration
- Analyse Farmer Input Data
- Manage crop & Vegetables database
- Manage Seeds & Fertilizers Database
- Monitor Recommendations
- Monitors orders
- Manage deliveries
- Generate reports
- Manage godown Details
- Maintain System

- Log out

Customer:

- Register/Login
- Search vegetables
- Place order
- Make Payment
- Track delivery
- Receive Order
- Give Feedback

Farmer:

- Register/Login
- Enter soil and crop data
- Get crop recommendations from agricultural Experts
- View crop details
- Search Recommended seeds and Fertilizers
- Place Order
- Make payment
- Track Delivery
- Give Feedback
- List vegetables for selling to a Godown
- Send Request to purchase
- Transport vegetables, if godown Manager agrees
- Receive payment

Godown Manager:

- Register/Log in
- View Request
- Check Storage availability
- Notify Farmer
- Receives and Stores Vegetables
- Make Payment
- Record crop details(Type, Quantity, Date)

- Monitor Crop condition(Temperature, Moisture)
- Update Storage Records

MIS Information Levels & Expected Benefits

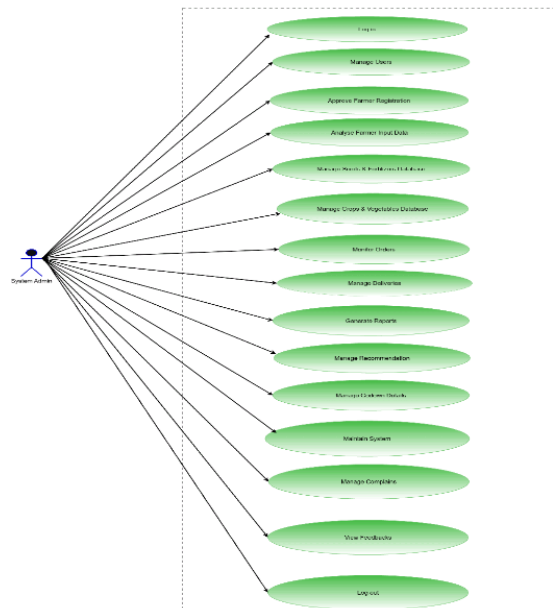
MIS Information Levels

- **Operational Information:**
 - Daily farmer inputs
 - Crop recommendations generated
- **Tactical Information:**
 - Monthly crop demand analysis
 - Regional crop trends
- **Strategic Information:**
 - Long-term agricultural planning
 - Government policy support

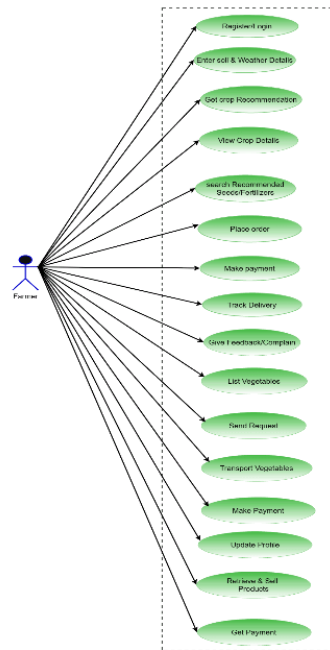
Expected Benefits

- Increased crop yield
- Better farming decisions
- Easy access to agricultural products
- Direct market access for farmers
- Improved income and productivity

Admin Use Case Diagram



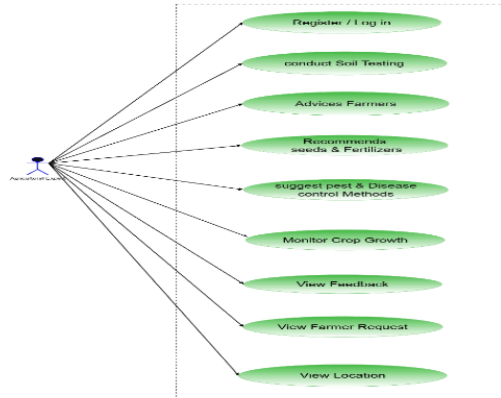
Farmer Use case Diagram



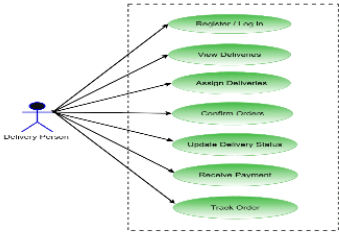
Customer Use Case Diagram



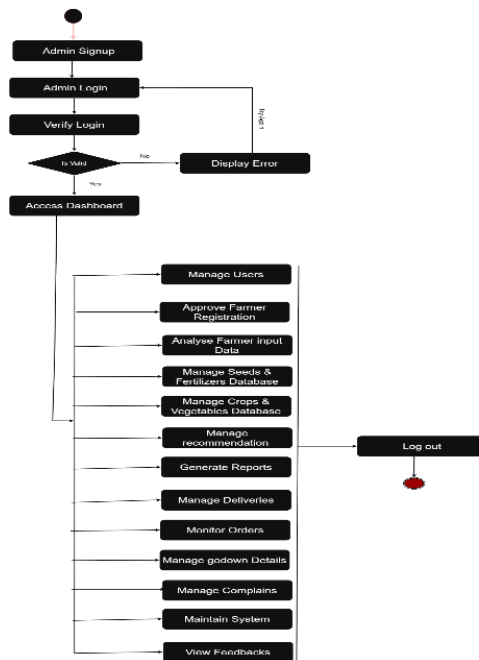
Agricultural Expert Use Case Diagram



Delivery Person Use Case Diagram



Admin Activity Diagram



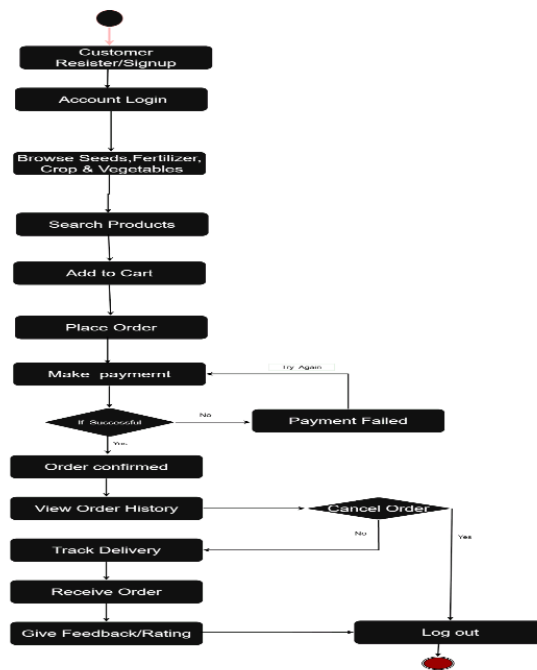
Delivery person Activity Diagram



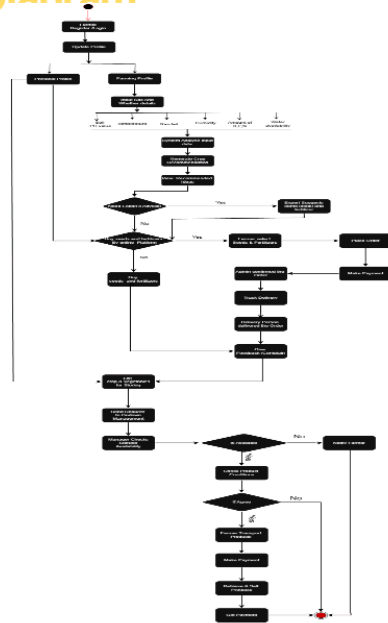
Agricultural Expert Activity Diagram



Customer Activity Diagram



Farmer Activity Diagram



Farmer Sequence Diagram

